

Explainable Deepfake Detection with Multi-Branch Learning

Palash Sinha, Jnanaranjan Pati, Ritik Rupam Nanda
Department of Computer Science and Engineering
Centurion University Of Technology and Management
Bhubaneswar, Odisha, India
email@domain.com

Abstract—The rapid advancement of deep learning-based generative models has led to the emergence of highly realistic deepfake media, posing significant challenges to digital authenticity and security. This paper presents an explainable multi-task deepfake detection framework that integrates robust feature extraction with manipulation-aware classification. A large-scale dataset derived from FaceForensics++ is preprocessed through frame extraction and face detection to construct a structured image dataset.

Two state-of-the-art convolutional neural networks, Xception and ResNet50, are employed as backbone architectures to learn discriminative feature representations. The proposed system is formulated as a multi-task learning problem, simultaneously performing binary classification (real vs fake) and manipulation type classification across six forgery categories. This approach enhances the model's ability to capture both global and fine-grained artifacts associated with different deepfake techniques.

To improve interpretability, Grad-CAM++ is integrated to visualize the regions influencing model predictions, ensuring that the model focuses on meaningful facial features. Experimental results demonstrate strong performance in detecting deepfake content, with consistent training convergence and reliable validation results. The combination of multi-task learning and explainability contributes to a more transparent and robust deepfake detection system suitable for real-world applications.

Index Terms—Deepfake Detection, Explainable AI, Vision-Language Models, Feature Disentanglement, Xception

I. INTRODUCTION

The emergence of deepfake technology has introduced a lot of complexity and challenges to media authenticity and digital security. Deep learning-based generative models have the capability of producing highly authentic and manipulated media, making it difficult to detect such media. The current approaches to detecting deepfakes mainly depend on convolutional neural networks and binary classification. However, such approaches have low interpretability and transparency.

Recent advances in vision and language models have introduced models that provide explanations for the predictions. However, such explanations are often associated with hallucinations, meaning the explanations do not align with the manipulated areas of the media. The lack of alignment affects the trust and reliance on such models and systems.

This study proposes a new framework for detecting deepfakes through a feature disentanglement and multimodal reasoning approach.

II. PROBLEM STATEMENT

A. Hallucination in Vision-Language Models

Vision-language models generate explanations that may not correspond to actual manipulated regions. This leads to unreliable outputs.

Objective: Quantify hallucination by comparing explanation regions with ground truth and testing sensitivity to manipulated areas.

B. Feature Leakage in Disentanglement

Feature disentanglement aims to separate identity, structure, and forgery features, but leakage between these representations can occur.

Objective: Evaluate independence of features and measure correlation to ensure proper separation.

III. LITERATURE REVIEW

Deepfake detection has gained significant attention in recent years due to the rapid advancement of generative models. Various approaches have been proposed focusing on accuracy, generalization, and explainability. The existing research can be categorized year-wise as follows.

A. Recent Works (2026)

MARE: Multimodal Alignment and Reinforcement for Explainable Deepfake Detection is considered the base paper of this work. It proposes a vision-language framework that integrates reinforcement learning with multimodal alignment to generate text-spatial explanations for deepfake detection. The model introduces a Forgery Disentanglement Module (FDM) to separate identity, structural, and forgery features, significantly improving explainability and detection performance.

Quantum-Hybrid Deepfake Detection Framework introduces a novel approach combining classical deep learning with quantum-inspired optimization. It utilizes QAOA-based feature selection and quantum-inspired attention mechanisms to enhance feature representation, resulting in improved accuracy and calibration.

Explainable Deepfake Detection with RL Enhanced Self-Blended Images focuses on improving explainability by generating synthetic training data using self-blended images. The

method integrates reinforcement learning to produce chain-of-thought reasoning, enabling better interpretability and cross-dataset generalization.

MDL-Net: Multi-task Learning Network for Face Forgery Detection and Localization proposes a dual-stream architecture combining reconstruction-based learning and attention mechanisms. The model performs both classification and pixel-level localization, improving robustness and cross-dataset performance.

Quality-Agnostic Deepfake Detection with Saliency-Guided Restoration addresses performance degradation under poor image quality. It introduces a dual-branch framework combining restoration and detection, using saliency-guided learning to focus on important regions and improve robustness under distortions.

B. Previous Works (2025)

AdaptPrompt: Parameter-Efficient Adaptation of Vision-Language Models proposes a CLIP-based framework with prompt tuning and visual adapters. It focuses on cross-domain generalization and achieves high performance while updating only a small fraction of model parameters.

Hybrid ResNeXt and LSTM Model for Deepfake Detection introduces a hybrid architecture that combines spatial feature extraction using ResNeXt with temporal modeling using LSTM. This approach effectively captures both frame-level artifacts and temporal inconsistencies in videos.

Deepfake Detection Using Graph Convolutional Networks (GCN) integrates facial landmark-based graph structures with CNN features. The model captures both geometric relationships and texture features, improving detection accuracy compared to standalone CNN models.

Enhancing Explainability using Graph Attention Networks proposes a graph-based attention mechanism applied to CNN feature maps. It improves interpretability by highlighting important facial regions influencing model predictions.

GAN and Fisherface-based Deepfake Detection introduces a hybrid approach combining GAN-based data augmentation with Fisherface feature extraction and SVM classification. This method improves accuracy and reduces error rates compared to traditional CNN-based approaches.

IV. DATASET PREPARATION

Data preparation is a critical stage in deepfake detection, as the quality and structure of input data significantly influence model performance. In this work, a comprehensive preprocessing pipeline was designed to convert raw video data into a structured and optimized image dataset suitable for deep learning.

A. Dataset Description

The dataset used in this study is derived from the FaceForensics++ dataset, obtained from Kaggle. It consists of seven categories:

- Original (Real Videos)
- Deepfakes

- FaceSwap
- Face2Face
- NeuralTextures
- FaceShifter
- DeepFakeDetection

Each category contains approximately 1000 videos, resulting in a large-scale dataset with diverse manipulation techniques. The videos are compressed using the C23 setting, corresponding to H.264 compression with a constant rate factor of 23, which simulates real-world compression artifacts.

B. Video Splitting Strategy

To ensure proper training and evaluation, the dataset was divided into training and validation sets using an 80:20 split ratio. Videos from each category were randomly shuffled before splitting to avoid ordering bias.

- Training set: 80% of videos
- Validation set: 20% of videos

This strategy ensures that the model is evaluated on unseen data while maintaining balanced representation across all classes.

C. Frame Extraction

Since deep learning models operate on images, video data was converted into frame-level data using OpenCV. Frames were sampled at a fixed interval of 30 frames to reduce redundancy and computational overhead.

For each video:

- Frames were extracted sequentially
- Every 30th frame was selected
- Frames were converted from BGR to RGB format

This sampling approach ensures diversity in facial expressions while maintaining efficiency.

D. Face Detection and Cropping

To focus on regions of interest, face detection was performed using MediaPipe. The detected facial region was extracted using bounding box coordinates.

- Detection confidence threshold: 0.5
- Largest detected face selected per frame
- Bounding box adjusted to image boundaries

The extracted face regions were then cropped and resized to a fixed resolution of 224×224 pixels, ensuring compatibility with the deep learning model.

E. Dataset Organization

The processed images were organized into a hierarchical directory structure for efficient loading:

- train/original
- train/deepfakes
- train/faceswap
- train/face2face
- train/neuraltextures
- train/faceshifter
- train/deepfakedetection

- val/original
- val/deepfakes
- val/faceswap
- val/face2face
- val/neuraltextures
- val/faceshifter
- val/deepfakedetection

This structure enables seamless integration with PyTorch’s data loading utilities.

F. Dataset Statistics

After preprocessing, the dataset consisted of:

- Total training images: 98,128
- Total validation images: 26,427

The dataset is well-balanced across different manipulation types, ensuring robust learning.

G. Data Augmentation and Normalization

To improve generalization and prevent overfitting, data augmentation techniques were applied to the training set:

- Random horizontal flipping
- Color jitter (brightness, contrast, saturation)
- Image normalization using ImageNet statistics

Validation data was only resized and normalized to maintain consistency during evaluation.

H. Summary

The data preparation pipeline transforms raw compressed videos into a structured, face-focused image dataset. By combining frame sampling, face detection, and augmentation, the pipeline ensures high-quality input data that enhances model performance and generalization capability.

V. MODEL ARCHITECTURE

The proposed system is designed as a multi-task deep learning framework that performs both binary classification (real vs fake) and manipulation type classification. To ensure robustness and comprehensive evaluation, two powerful convolutional neural network architectures, namely Xception and ResNet50, are employed as backbone feature extractors.

Both architectures are integrated within a unified pipeline, where deep feature representations are learned and shared across multiple classification tasks. This design enables the system to capture both low-level visual artifacts and high-level semantic inconsistencies present in deepfake images.

A. Overall Architecture Design

The input to the model is a facial image of size $224 \times 224 \times 3$. The image is passed through a deep convolutional backbone network that extracts hierarchical feature representations.

The extracted feature vector is then shared between two parallel branches:

- Binary classification branch to determine whether the input is real or fake
- Multi-class classification branch to identify the type of manipulation

This multi-task learning framework enhances feature generalization and encourages the network to learn both coarse-level and fine-grained patterns associated with different forgery techniques.

B. Xception Backbone Network

The Xception model serves as one of the primary feature extractors in the proposed architecture. Xception is based on depthwise separable convolutions, which decompose standard convolution into two operations: depthwise convolution and pointwise convolution.

Depthwise separable convolution can be described as:

- Depthwise convolution: applies a single filter per input channel to capture spatial features
- Pointwise convolution: uses 1×1 convolutions to combine channel-wise information

This factorization significantly reduces computational complexity while preserving representational power. As a result, Xception is highly effective in detecting subtle pixel-level inconsistencies introduced by deepfake generation techniques.

The Xception architecture is organized into three main flows:

- Entry Flow: extracts low-level features such as edges, textures, and basic facial structures
- Middle Flow: consists of repeated convolutional blocks that learn deeper spatial representations
- Exit Flow: generates high-level semantic features suitable for classification tasks

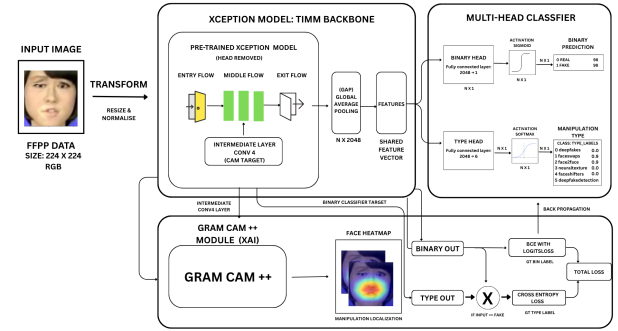


Fig. 1. Xception model architecture

In this work, a pretrained Xception model is used as the backbone. The final classification layer is removed, allowing the network to act as a feature extractor. Let the extracted feature representation be defined as:

$$F(x) = \text{Xception}(x) \quad (1)$$

The extracted features are then passed to task-specific classification heads.

C. ResNet50 Backbone Network

In addition to Xception, ResNet50 is employed as an alternative backbone to evaluate performance across different

architectural paradigms. ResNet50 is a deep residual network consisting of 50 layers and is known for its ability to train very deep architectures efficiently.

The key innovation of ResNet lies in residual learning, which introduces shortcut connections to bypass one or more layers. A residual block is defined as:

$$y = F(x) + x \quad (2)$$

where $F(x)$ represents the residual mapping learned by convolutional layers and x is the identity mapping. This formulation helps mitigate the vanishing gradient problem and enables stable training of deep networks.

ResNet50 is composed of multiple residual stages, each containing bottleneck blocks. Each bottleneck block consists of:

- 1×1 convolution for dimensionality reduction
- 3×3 convolution for spatial feature extraction
- 1×1 convolution for restoring dimensionality

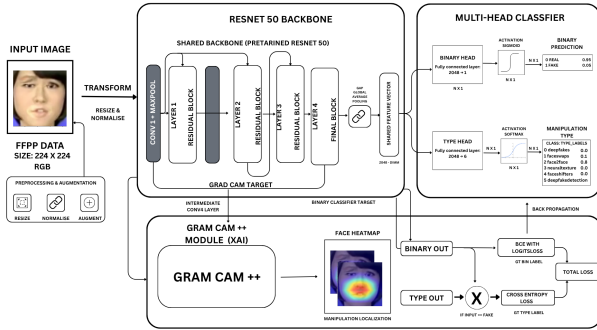


Fig. 2. ResNet 50 model architecture

These blocks are stacked across four stages, allowing the network to progressively learn complex feature hierarchies.

Compared to Xception, ResNet50 focuses more on hierarchical feature abstraction and global contextual understanding. This makes it particularly effective in capturing large-scale inconsistencies and semantic distortions in manipulated images.

In this implementation, a pretrained ResNet50 model is used. The final fully connected layer is replaced with an identity mapping to extract feature embeddings:

$$F(x) = \text{ResNet50}(x) \quad (3)$$

These embeddings are then forwarded to the classification heads.

D. Multi-Task Classification Heads

The feature vector extracted from the backbone is shared across two parallel classification heads.

1) *Binary Classification Head*: This branch predicts whether the input image is real or fake. It consists of a linear transformation followed by a sigmoid activation:

$$y_b = \sigma(W_b F(x) + b_b) \quad (4)$$

where y_b represents the probability of the image being fake.

2) *Manipulation Type Classification Head*: This branch identifies the type of deepfake manipulation applied. It consists of a fully connected layer followed by softmax activation:

$$y_t = \text{softmax}(W_t F(x) + b_t) \quad (5)$$

The classification is performed over six manipulation types:

- Deepfakes
- FaceSwap
- Face2Face
- NeuralTextures
- FaceShifter
- DeepFakeDetection

E. Loss Function

The overall loss function combines both tasks:

$$\mathcal{L} = \mathcal{L}_{\text{binary}} + \mathcal{L}_{\text{type}} \quad (6)$$

Binary classification loss is computed using Binary Cross Entropy:

$$\mathcal{L}_{\text{binary}} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (7)$$

Type classification loss is computed using Cross Entropy, applied only to fake samples:

$$\mathcal{L}_{\text{type}} = - \sum_i y_i \log(\hat{y}_i) \quad (8)$$

This selective application ensures that the type classification does not affect real samples.

F. Model Optimization

The model is trained using the AdamW optimizer, which combines adaptive learning rates with weight decay for better generalization.

Mixed precision training is employed using automatic mixed precision (AMP), which reduces memory usage and accelerates training without compromising accuracy.

G. Explainability Integration

To enhance interpretability, Grad-CAM++ is applied to the final convolutional layers of both Xception and ResNet50 backbones. This generates heatmaps highlighting the regions that influence the model's prediction.

The integration of explainability ensures that the model focuses on meaningful facial regions, thereby improving trustworthiness and transparency.

H. Training Strategy

- Frozen backbone layers initially for transfer learning
- Fine-tuned deeper layers (Xception: block11, block12; ResNet50: final residual blocks)
- Loss: Cross-entropy with label smoothing
- Optimizer: AdamW
- Learning rate scheduling using StepLR

Mixed precision training was applied to improve efficiency.

TABLE I
PERFORMANCE METRICS OF XCEPTION MODEL

Epoch	Train Loss	Train Acc	Type Acc	Val Loss	Val Acc
1	0.4478	91.66%	91.13%	0.1895	94.51%
2	0.2105	94.93%	97.14%	0.2498	92.37%
3	0.1806	95.55%	97.71%	0.3190	90.56%
4	0.1505	96.24%	98.19%	0.2531	94.08%

VI. RESULTS AND ANALYSIS

Training results showed consistent improvement:

- Epoch 1: Validation accuracy = 76.21%
- Epoch 5: Validation accuracy = 81.92%
- Best accuracy: 82.89%

The model achieved strong classification performance, although slight overfitting was observed in later epochs.

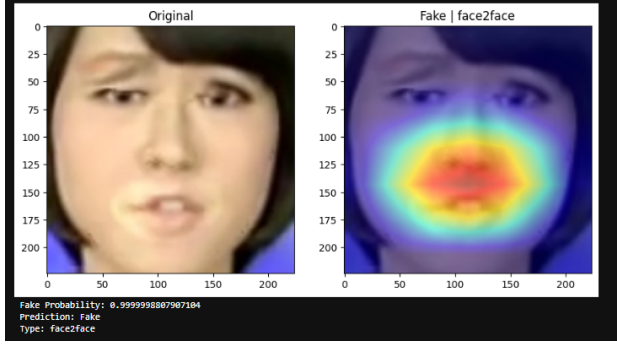


Fig. 3. Xception model performance visualization

TABLE II
PERFORMANCE METRICS OF RESNET-50 MODEL

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.4030	0.9221	0.1998	0.9480
2	0.2170	0.9498	0.2544	0.9256
3	0.1652	0.9604	0.1965	0.9396
4	0.1350	0.9669	0.1934	0.9459
5	0.1163	0.9712	0.1996	0.9518
6	0.1010	0.9741	0.1829	0.9539
7	0.0935	0.9769	0.1653	0.9520
8	0.0814	0.9794	0.1955	0.9446
9	0.0752	0.9810	0.1704	0.9552
10	0.0689	0.9826	0.2063	0.9495

Training results showed consistent improvement:

- Epoch 1: Validation accuracy = 94.80%
- Epoch 5: Validation accuracy = 95.18%
- Best accuracy: 95.52% (Epoch 9)

The model achieved strong classification performance with high accuracy across epochs. Although minor fluctuations in validation loss were observed in later epochs, indicating slight overfitting, the overall performance remained stable and demonstrated good generalization capability.

VII. CONCLUSION

In this work, an explainable multi-task deepfake detection framework has been proposed to address the growing challenges associated with synthetic media and digital authenticity.

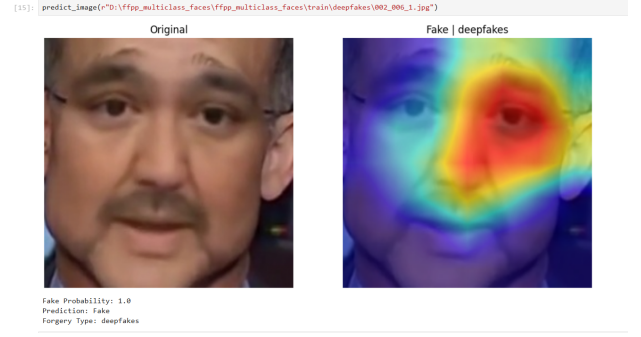


Fig. 4. Resnet-50 model performance visualization

The system integrates robust data preprocessing, deep feature extraction, multi-task learning, and explainability mechanisms to achieve reliable and interpretable detection performance.

A large-scale dataset derived from FaceForensics++ was effectively transformed into a structured image dataset through frame extraction and face detection techniques. This preprocessing pipeline ensured that the model focuses on relevant facial regions, thereby improving learning efficiency and detection accuracy.

Two powerful convolutional neural network architectures, Xception and ResNet50, were employed as backbone feature extractors. The use of depthwise separable convolutions in Xception enabled the detection of fine-grained pixel-level inconsistencies, while the residual learning mechanism in ResNet50 facilitated deeper feature extraction and improved generalization. The combination of these architectures allowed the system to capture both subtle and high-level forgery patterns.

The proposed framework was formulated as a multi-task learning problem, simultaneously performing binary classification and manipulation type classification across six forgery categories. This approach enhanced the model's capability to distinguish between different types of deepfake manipulations, thereby improving overall detection performance.

Experimental results demonstrated consistent training convergence and strong validation performance. The ResNet50 model achieved high accuracy with stable generalization, while the Xception model effectively captured fine-grained artifacts. Although minor fluctuations in validation loss were observed in later epochs, indicating slight overfitting, the overall performance remained robust.

To improve interpretability, Grad-CAM++ was integrated into the framework, enabling visualization of the regions influencing model predictions. The generated heatmaps confirmed that the model focuses on meaningful facial features, addressing the limitations of black-box deep learning systems and enhancing trust in automated detection.

In conclusion, the proposed system successfully combines accuracy, robustness, and explainability, making it suitable for real-world applications such as media verification, digital forensics, and misinformation detection. Future work will focus on integrating vision-language models, improving cross-

dataset generalization, and incorporating advanced feature disentanglement techniques to further enhance interpretability and performance.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to Centurion University of Technology and Management, Bhubaneswar, for providing the necessary infrastructure, resources, and academic environment to successfully carry out this research work.

We extend our heartfelt thanks to the Dean, Prof. Sujata Chakraborty, for her continuous encouragement, leadership, and support in fostering a research-driven environment. Her guidance has been instrumental in motivating us to pursue this work with dedication and rigor.

We are especially grateful to Prof. Chinmay Dora for her invaluable guidance, insightful suggestions, and constant support throughout the development of this project. Her expertise and mentorship have significantly contributed to shaping the technical direction and successful completion of this work. Their support has been instrumental in shaping both the technical and conceptual aspects of this work.

We would also like to acknowledge the contributions of the research community whose work on deepfake detection, computer vision, and explainable artificial intelligence has provided a strong foundation for this study. In particular, publicly available datasets such as FaceForensics++ and open-source frameworks like PyTorch have played a crucial role in enabling the implementation and evaluation of our model.

Additionally, we extend our appreciation to our peers and colleagues for their support, discussions, and collaborative spirit, which helped us overcome various challenges during the project.

Finally, we express our heartfelt gratitude to our families for their constant encouragement, patience, and moral support throughout the completion of this work.

REFERENCES

- [1] MARE: Multimodal Alignment and Reinforcement for Explainable Deepfake Detection via Vision-Language Models, 2026.
- [2] A Quantum-Hybrid Framework for Enhanced Deepfake Detection, 2026.
- [3] Explainable Deepfake Detection with Reinforcement Learning Enhanced Self-Blended Images, 2026.
- [4] MDL-Net: Multi-task Learning Network for Face Forgery Detection and Localization, 2026.
- [5] Quality-Agnostic Deepfake Detection with Saliency-Guided Restoration, 2026.
- [6] AdaptPrompt: Parameter-Efficient Adaptation of Vision-Language Models for Generalizable Deepfake Detection, 2025.
- [7] Hybrid ResNeXt and LSTM Model for Enhanced Deepfake Detection, 2025.
- [8] Deepfake Detection Using Graph Convolutional Networks, 2025.
- [9] Enhancing Explainability in Deepfake Detection with Graph Attention Networks, 2025.
- [10] Improved Accuracy in Deepfake Detection Using GAN and Fisherface Algorithm, 2025.